

Response to Reviewer 3

We thank the reviewer for reviewing our work and giving thoughtful and constructive comments. We have made several changes that you have suggested, and we detail specifically how we have addressed your comments below.

Comment 1

My main concern with your current paper is mostly an inbreeding estimation paper and not an economics paper. As such, it seems it was targeted to the wrong journal. That said, you appear to have developed some expertise in the area, so perhaps more economic work in what is an unfilled area in economics will come from future work.

Reply: Thank you for pointing this out. In our last draft, there were parts of the paper that misled readers into thinking this paper was about estimating inbreeding. In fact, the goal of the paper is to analyze the strategic actions of dairy breeders and how genomic testing has changed their decisions. In our new conceptual model we explain the extent to which increased inbreeding rates are a consequence of intense competition among bull breeders to provide the market with the highest quality bulls, since their profits depend on them. We have also added a new paragraph in page 15 explaining how the inbreeding rate is calculated by the NNAB using extensive progeny data:

The NAAB lists also contain information about lineage, which we use to track descendants of a bull. Every animal born in the USA and Canada is assigned a unique ID which includes information about its breed, country of birth. The NAAB data includes these IDs for the bull and both of its parents, allowing us to build a genetic network of relationships from the data. Inbreeding is commonly measured using the **inbreeding coefficient** (F_X) (Sewall Wright, “Coefficients of Inbreeding and Relationship” (1922) 56(645) The American Naturalist 330), defined as the probability that two genes in an individual were both descended from a single gene in an ancestor.

$$F_X = \sum_{ca=1}^K \left(\frac{1}{2}\right)^{n_1+n_2+1} (1 + F_{ca}) \quad (1)$$

Where ca denotes the common ancestor, n_j is the number of generations separating the common ancestor from the sire (dam), and F_{ca} is the inbreeding of the common ancestor. This value can be calculated before the birth of an animal, since it depends on the inbreeding coefficients of its parents. The inbreeding coefficient is reported when a newborn animal is registered in its database.

Comment 2

Does figure 2 have a source? Also, what are units on inbreeding rate and how is it calculated? You use this inbreeding term quite a bit and it needs more explanation and definition. Same

for TPI on units and how calc in fig 2, though not as important for your paper as inbreeding rate.

Reply: A source was added to Figure 2. Inbreeding is measured with the **Inbreeding coefficient**, which is equal to the probability that two alleles at the same locus are identical by descent, that is, the probability of inheriting the alleles from their dam and sire. In practice, the CDCB calculates this number using **progeny** data. Consequently, the inbreeding rate is bounded between zero (absolutely no relation) and 1 (a calf born from parents that are siblings and thus genetically equal to them).

The Total Productivity Index is a selection index, a weighted average of PTA values that assigns more weight to specific traits over others to summarize information into a single value per animal, which can be used to rank bulls from highest to lowest.

An explanation of both measures has been included in the text to avoid further confusion. On pg. 14, we explain the inbreeding coefficient and the TPI index, which is explained in the footnote on page 3.

Comment 3

Page 6. Your back-of-the-envelope cost of inbreeding that you hang your hat on and note in the abstract of your paper is at much at \$6.7 billion. What is this cost? You mention Holstein USA but I do not see this reference in your reference list. When I review the publication at: https://www.holsteinusa.com/pdf/Interpreting_and_Utilizing_Haplotype_Information_1218.pdf (which is in your references), I see no mention of the cost you calculate, but rather information about specific DNA sequences that are of concern. Then further, you state that the added milk productivity resulted in a net gain that exceeded this \$6.7 billion cost. So, net, genetic testing to date has been a huge benefit to the dairy industry if I am interpreting your claims correctly. So, why are you anchoring on this “cost” measure of \$6.7 billion.

Reply: Thank you for this comment. Upon reflection, we have realized that this paper cannot do an effective cost-benefit analysis. Instead, the main goal of the paper is to analyze the actions of dairy breeders following the introduction of genomic testing. In this new revision, we have de-emphasized the costs that we calculate here. We incorporate a discount rate and a more detailed explanation of costs under two distinct sets of assumptions. A more detailed appraisal of costs and benefits would require a re-estimation of the Net Merit coefficient to reflect changes in both inbreeding and trait weights, but such a calculation is beyond the scope of this article and merits an independent paper by itself.

Comment 4

How much does genetic testing cost? Seems we need to know that to also assess net value.

Reply: A single genetic test costs \$35. Source: <https://vgl.ucdavis.edu/test/parentage-genetic-marker-report-cattle>. However, as we explained in the comment above, we no longer frame the cost=benefit analysis as our main contribution.

Comment 5

Your premise that inbreeding is a problem because it “may have hidden costs” and not that it has been demonstrated to be an economic problem. That is, you rely on conjecture that inbreeding will result in major disease or other genetic concerns that lead I guess to important economic consequences because some traits are being ignored or are not exhibited sufficiently due to inbreeding. While this may be true from a genetics perspective, it seems like conjecture on your part at this point and has no economic assessment in your paper. The paper by Littledon (2024) <https://www.rrcij.com/open-access/the-role-of-genomics-in-enhancing-dairy-cattle-productivity-and-health.pdf> indicates genomics testing is enabling selecting for milk yield, feed efficiency, and detection of inherited deficiencies and mutations – essentially a more productive and healthy dairy herd. So, has inbreeding become problematic or do we know when it is a problem? What is the rate of inbreeding at which it is a problem? Are we at a critical level economically yet or are we a long ways away?

Reply: Thank you for this comment. In this revision, we have emphasized that the costs of inbreeding are from i) inbreeding depression which manifests immediately and ii) lost welfare from keeping the level of genetic diversity lower than would be socially optimal. When there is less genetic diversity, there is less scope for genetic improvement, as having the whole gene pool made up of essentially the same animal would mean there are no more improvements to make.

This point is now made on pg. 7:

If inbreeding continues to increase in the industry, even more damages might occur. Studies already estimate that the Holstein breed has an unsustainable level of inbreeding, having an “effective population” of between 43 and 66; it is commonly assumed that 50 is a critical level under which the breed will lose resilience (Bayode O Makanjuola and others, “Effect of genomic selection on rate of inbreeding and coancestry and effective population size of Holstein and Jersey cattle populations” (2020) 103(6) Journal of Dairy Science 5183). Moreover, the lower the genetic diversity, the greater the risk there is of unknown recessive traits causing significant damage to genetic lines (JB Cole and PM VanRaden, “Use of haplotypes to estimate Mendelian sampling effects and selection limits” (2011) 128(6) Journal of Animal Breeding and Genetics 446; Emmanuel A Lozada-Soto and others, “Effects of germplasm exchange strategies on genetic gain, homozygosity, and genetic diversity in dairy stud populations: A simulation study” (2024) 107(12) Journal of Dairy Science 11149). At its worst, the population can reach the maximum possible increase in genetic trait values leading to a stalling of genetic improvement. Future work in economics should seek to understand how policy might be used to address market frictions and incentivize breeders to outcross to ameliorate the impacts of inbreeding.

Comment 6

Dairy producers are in business for the long run, so why would they act irrationally and select sires for only short-run gains and ignore disease resistance? Do you have any evidence that they actually do this? That is, breeders must provide genetics demanded by producers for long-run decisions. In the Bt corn case you discuss it was the seed companies that required refuge being planted so they could ensure the technology would be reliable in accomplishing what it claimed for the long run. This was not a federal government mandate. So, why would the same not occur in dairy genetics? You imply federal intervention is warranted because of an externality but you have not demonstrated such an externality exists or its magnitude.

Reply: In our new revision, our conceptual model shows that it is individually rational for a dairy breeder to inbreed more than is socially optimal because of the competitive pressures of the industry. Also, outcrossing provides both private and public value to firms but they are only incentivized to care about the private value, which means they will choose outcrossing less than would be optimal to maximize welfare.

Comment 7

Do dairy producers not also conduct genetic tests increasingly on young heifers for deciding which ones to retain? If so, wouldn't they also match that genetic info with sires for better complementarity of overall traits?

Reply: Dairy farmers can and do genetically test their female cattle to determine which cow they should mate with each bull on the market. However, it is unclear whether dairy farmers know the inbreeding coefficient of their choices or factor this into their decisions. This points to an information asymmetry issue in the industry. See John B Cole, "Perspective: Can we actually do anything about inbreeding?" (2024) 107(2) Journal of Dairy Science 643 (Publisher: Elsevier) for a discussion on how dairy farmers view inbreeding in the industry. In general, dairy farmers do not keep male cattle within their herds. Most of the semen sold in the market is "sexed", meaning that it has a higher probability of giving birth to a female by artificially decreasing the number of Y-chromosomes (see <https://www.absglobal.com/uk/what-is-sexed-semen-in-cattle/>). If, by pure chance, a male is born, the farmer is often contractually required to return the animal to the genetics company in exchange for more samples or to sell it for its beef once it is old enough.

A different breeding strategy involves embryo transfer, in which cows are stimulated to produce a large number of eggs, which are artificially inseminated and transferred to the original cow or a third cow. This strategy also allows for sex-sorting with a very high accuracy. In other words, it is highly unlikely that a dairy farmer keeps bulls in their herds.

Comment 8

Figure 4. Why has the number of bulls in the market gone up so rapidly from 2010 to 2015? How does this mesh with your inbreeding measures? Are many of these bulls closely inter-related with each other? At surface, it would seem more bulls would imply more genetic

diversity in the market, but perhaps not? Might deserve some explanation.

Reply: Thank for this good question. The reason for this increase is that genomic testing allows companies to market bulls before they are sexually mature instead of waiting 5 years for data on their offspring. This allows them to expand their bull offerings. However, this does not necessarily imply that the bulls are not related to each other. In fact, our analysis shows that the average inbreeding coefficient increased after more bulls are introduced, likely because these bulls are even more interrelated than the offerings before genomics. This is itself a consequence of the fact that breeders are more likely to inbreed after genomic testing because it increased the value of inbreeding over outcrossing (see our conceptual model).

Comment 9

Page 14 you state “Moreover, bulls that are not genomically tested are more likely to have better quality traits and be more widely used.” Why would bulls that are not tested be more likely to have better quality traits?

Reply: That sentence was written in error and has been removed from the text.

Comment 10

Page 16, Figure 5 appears in the text, not the Appendix as stated here.

Reply: The sentence was corrected to exclude references to the appendix.

References 3

- Wright S, “Coefficients of Inbreeding and Relationship” (1922) 56(645) The American Naturalist 330.
- Makanjuola BO and others, “Effect of genomic selection on rate of inbreeding and coancestry and effective population size of Holstein and Jersey cattle populations” (2020) 103(6) Journal of Dairy Science 5183.
- Cole JB and VanRaden PM, “Use of haplotypes to estimate Mendelian sampling effects and selection limits” (2011) 128(6) Journal of Animal Breeding and Genetics 446.
- Lozada-Soto EA and others, “Effects of germplasm exchange strategies on genetic gain, homozygosity, and genetic diversity in dairy stud populations: A simulation study” (2024) 107(12) Journal of Dairy Science 11149.
- Cole JB, “Perspective: Can we actually do anything about inbreeding?” (2024) 107(2) Journal of Dairy Science 643 (Publisher: Elsevier).